trymtex Showcase

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1 Theorems

Definition 1. Norm

A norm on a vector space V is a map $\|\cdot\| : V \to \mathbb{R}$.

Lemma 1. Nonnegativity

For all $x \in \mathbb{R}$, $x^2 \geq 0$.

Theorem 1. Pythagorean theorem

For a right triangle with legs a and b and hypotenuse c, we have $a^2 + b^2 = c^2$.

Problem 1. (Right triangles)
Find the hypotenuse of a right triangle with legs 3 and 4.

Exercise 1. (Squares)
Show that $x^2 \geq 0$ for every real number x.

Solution 1. (Squares)
If $x \geq 0$, then $x^2 \geq 0$. If $x < 0$, then $-x > 0$ and $x^2 = (-x)^2 \geq 0$.

Proof 1. (Pythagorean theorem)
Left as an exercise to the reader.

Remark 1. (Remark title)
This is a remark.

Example 1. (Example title)
This is an example.

Hint 1. Hint title
This is a hint.

This is a checkpoint.

2 Boxes

Filing
Content.

Railing
Content.

FlagFlag Content.

3 Code

```
def square(n: int) -> int:
    return n * n
```

4 Plots

